Introduction

My region of choice is the eastern Agulhas Bank region (34 - 37ºS; 22 - 26ºE). I chose the eastern section mainly to limit the size of my grid and because I thought the angle of the continental shelf in this region was interesting. The Agulhas Region is characterized by its strong current and extensive continental shelf along the southern coast of South Africa. This creates a complex system of drivers for productivity including shelf currents, multiple coastal upwellings and interactions with the main Agulhas Current (Jacobs et al., 2022).

Data Sets

Bathymetry Data

Bathymetric data were obtained from the Global Multi-Resolution Topography (GMRT) synthesis (Ryan et al., 2009), which provides global seafloor elevation data derived from integrated multibeam sonar and satellite altimetry observations. A subset of the Agulhas region (34–37°S; 22–26°E) was extracted using the GMRT MapTool (https://www.gmrt.org/). The dataset is used to characterise seafloor topography within the study region and to provide spatial context for oceanographic processes, including variations in chlorophyll concentration. The data was downloaded as a gridded NetCDF file, compatible with analysis in Python using the *xarray* library. The dataset consists of a two-dimensional spatial grid defined by longitude and latitude dimensions (*nx, ny*), with a total of *nx* × *ny* grid points (stored initially as a flattened array of size *xysize*). The data are provided on a regular grid in a cylindrical equidistant projection. Spatial resolution is determined by the grid spacing parameter, with coverage limited to the selected Agulhas region. The dataset includes the following key variables:

- *x_range, y_range:* Minimum and maximum longitude and latitude values
- *z_range:* Range of elevation values
- *spacing:* Grid resolution in the x and y directions
- *dimension:* Number of grid points in each spatial direction
- *z*: Seafloor elevation (m), with negative values representing depths below sea level

From these variables, latitude and longitude arrays are reconstructed and the depth field is reshaped into a two-dimensional grid for analysis.

File size: 12.2 MB

Source: Gmrt.org. (2025). *GMRT MapTool*. [online] Available at: https://www.gmrt.org/GMRTMapTool/.

Chlorophyll Data

Chlorophyll-a concentration data were obtained from the Ocean Colour Climate Change Initiative (OC-CCI) dataset (https://climate.esa.int/en/projects/ocean-colour/) developed by the European Space Agency (ESA) (Valente et al., 2016). This dataset provides a long-term, globally consistent record of ocean colour derived from merged satellite observations, including SeaWiFS, MODIS-Aqua, MERIS, VIIRS, and OLCI. Inter-sensor differences are minimised through bias correction and band-shifting techniques to ensure consistency across the time series. The dataset is used to quantify surface chlorophyll-a concentrations as a proxy for phytoplankton biomass, enabling the analysis of spatial patterns and seasonal variability in ocean productivity. The data are provided in NetCDF format and follow CF-1.7 conventions, making them suitable for analysis using the Python library *xarray*. The dataset consists of a three-dimensional grid with dimensions of 12 (time) × 4320 (latitude) × 8640 (longitude), representing a global dataset with high spatial resolution. The dataset represents monthly means, with 12 time steps corresponding to the seasonal cycle spanning September 1997 to August 1998. The data are provided on a global latitude–longitude grid with a spatial resolution of approximately 4 km at the equator. The dataset includes the following key variables:

- chlor_a: Chlorophyll-a concentration (mg m^{-3})
- lat, lon: Latitude and longitude coordinates (degrees)
- time: Monthly climatological time steps
- crs: Coordinate reference system information

File size: 1.79 GB

Source: Ocean Colour Climate Change Initiative dataset, Version <1.7>, European Space Agency, available online at http://www.esa-oceancolour-cci.org.

Figures

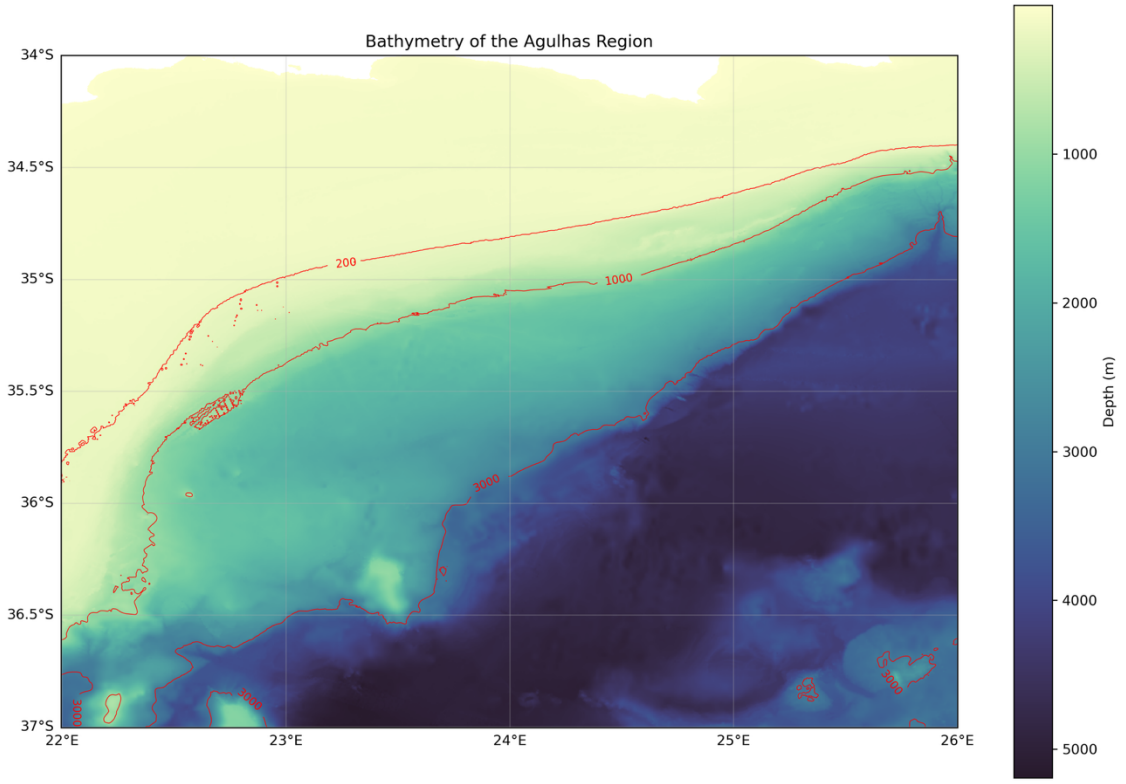

Figure 1: Map of the Agulhas Region (34-37ºS; 22-26ºE) showing bathymetry as depth below sea level in metres. The Agulhas Bank is above the 200m contour with the continental shelf sloping down towards the 3000m contour line.

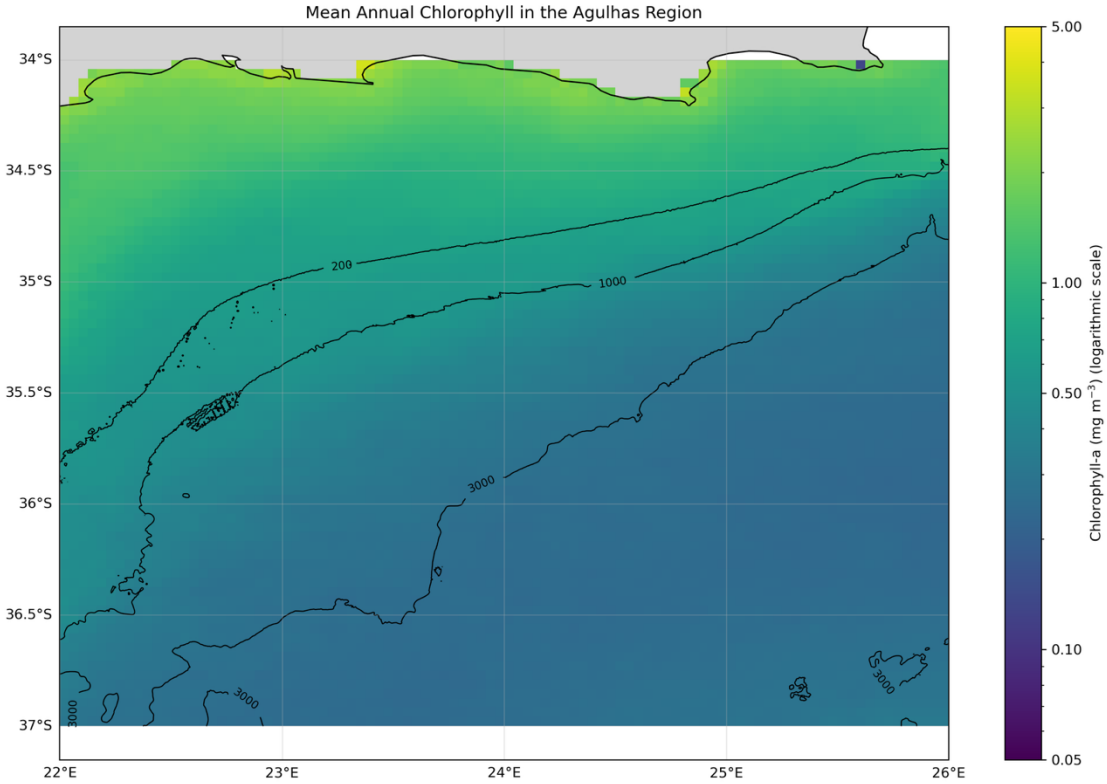

Figure 2: Map of mean annual chlorophyll concentration in the Agulhas Region (34-37ºS; 22-26ºE). To show the influence of bathymetry, the map is overlain with depth contours to show the shape of the Agulhas Bank. Chlorophyll concentration is plotted on a log scale.

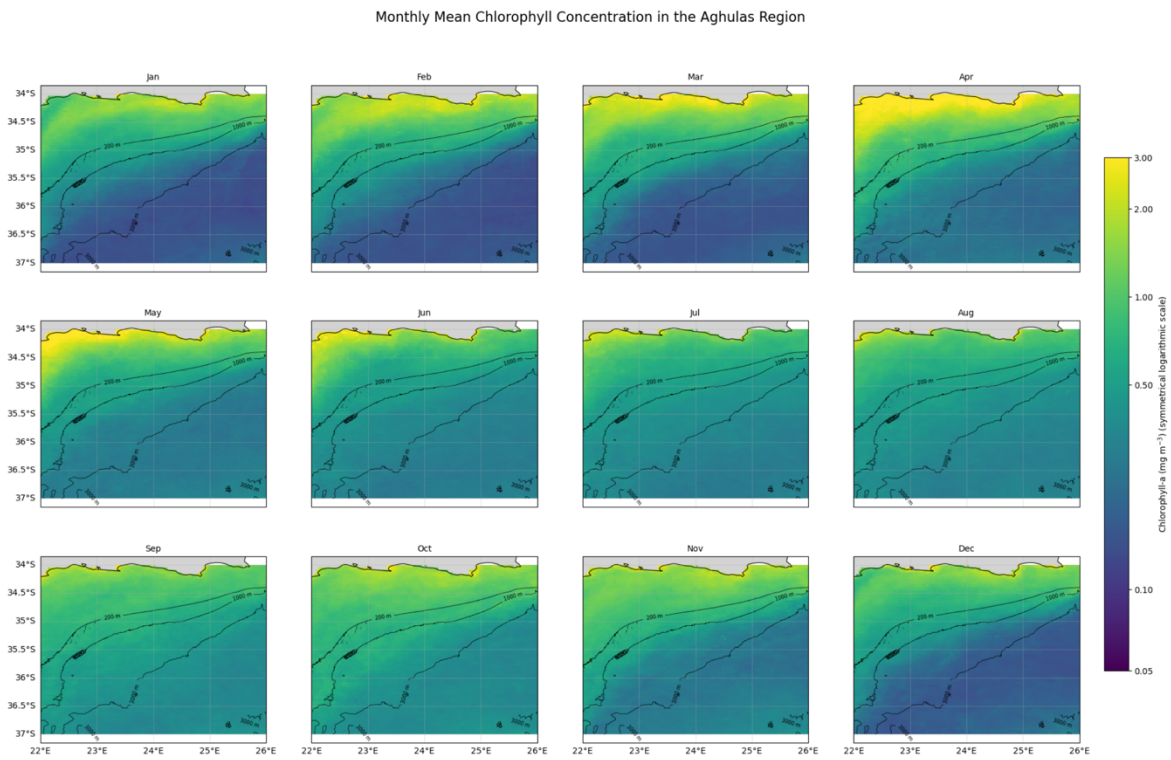

Figure 3: Monthly mean chlorophyll concentrations in the Agulhas Region (34-37ºS; 22-26ºE). To show the influence of bathymetry, the map is overlain with depth contours to show the shape of the Agulhas Bank. Concentrations peak in the austral autumn months. Chlorophyll concentration is plotted on a symmetric log scale.

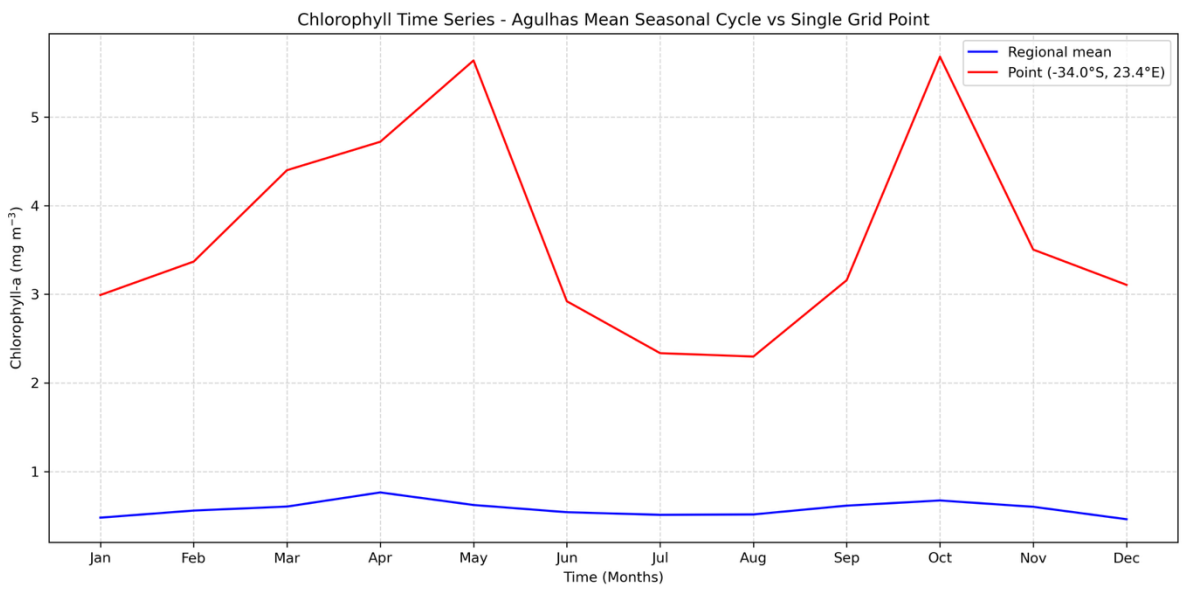

Figure 4: Time series of the mean seasonal cycle of chlorophyll concentration in the Agulhas Region compared to the mean seasonal cycle at the point of highest mean chlorophyll. Peaks in concentration occur in austral autumn and spring.

References

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